Nexum Suum – Expansion Tank Troubleshooting SOP

■ Page1: Pressure Fluctuation, Ruptured Bladder, and Thermal Over-pressure

If System Pressure Rises Rapidly When Heated:

→ Check Expansion Tank Pre-Charge – May be low or lost.
→ Verify Tank Isolation Valve is Open (if present).
→ Ensure Tank Sizing Matches Boiler or Loop Volume.

If Tank Appears Flooded or Heavy:

→ Test Schrader Valve for Air – No air = ruptured bladder.
→ Drain to Confirm If Waterlogged – Indicates diaphragm failure.

If Water is Discharging from Relief Valve:

→ Inspect Expansion Tank Charge and Volume First.
→ Check for Overfilled Make-Up Water Valve or Failed Fill Regulator.

If Tank is Corroded or Rusting Externally:

→ Evaluate Air Vent and pH Control in Loop.
→ Replace if Signs of Pitting, Leaks, or Pressure Failures Found.

Nexum Suum – Expansion Tank Troubleshooting SOP (Cont'd)

■ Page2: Symptom-to-Action Mapping+Real-World Thresholds

Symptom	Action
System Overpressure	Verify expansion tank air charge, size, and isolation valve.
Tank Flooded or Waterlogged	Check bladder integrity, test Schrader valve, drain tank.
Relief Valve Discharge	Inspect expansion tank, fill regulator, pressure controls.
Rust or Corrosion on Tank	Review water chemistry, ventilation, and pressure cycling.

■ Operating Thresholds:

→ Expansion Tank Pre-Charge: 12–25 psi (match system static fill pressure)
→ Acceptance Volume: Match expansion of entire loop (calculate via water volume × temp delta)
→ System Static Pressure (Cold): 12–18 psi typical for low-rise buildings
→ Max System Pressure (Hot): ≤30 psi (standard relief setting for hydronic boilers) Bladder Life
→ Expectancy: 5–10 years depending on cycling
→ Pressure Relief Valve Setting: 30 psi (standard), 50–150 psi for commercial Check pH: 8.5–10.5 for
→ closed-loop heating
→ Inspect 2×/year or during seasonal startup/shutdown